Gamma & Beta RNG

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Why Acceptance-Rejection? And How Does It Work?

The Gamma distribution has no closed-form inverse CDF, so direct sampling is impossible. Instead we use the Acceptance-Rejection method introduced by von Neumann (1951).

The idea: find a simple envelope $g(x)$ that sits above the target everywhere:

$$f(x) \leq c \cdot g(x) \quad \text{for all } x, \qquad \text{where } f(x) \propto x^{a-1} e^{-x}, \quad x \geq 0$$

Algorithm:

1. Sample a proposal $X \sim g(x)$ and draw $U \sim \text{Uniform}(0, 1)$ independently
2. Accept X as a draw from $f(x)$ if $U \leq \dfrac{f(X)}{c \cdot g(X)}$, otherwise reject and repeat

Why it works: the acceptance probability at any x is proportional to $f(x)/g(x)$. Tight regions are accepted often. Loose regions are rejected often. Accepted samples follow $f(x)$ exactly.

Efficiency: expected proposals per accepted sample = $c \int g(x)\, dx \geq 1$

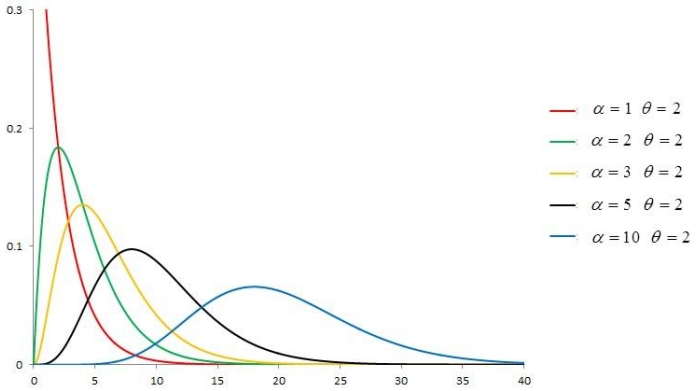
$$\text{Tighter envelope} \implies \text{smaller } c \implies \text{fewer wasted proposals}$$

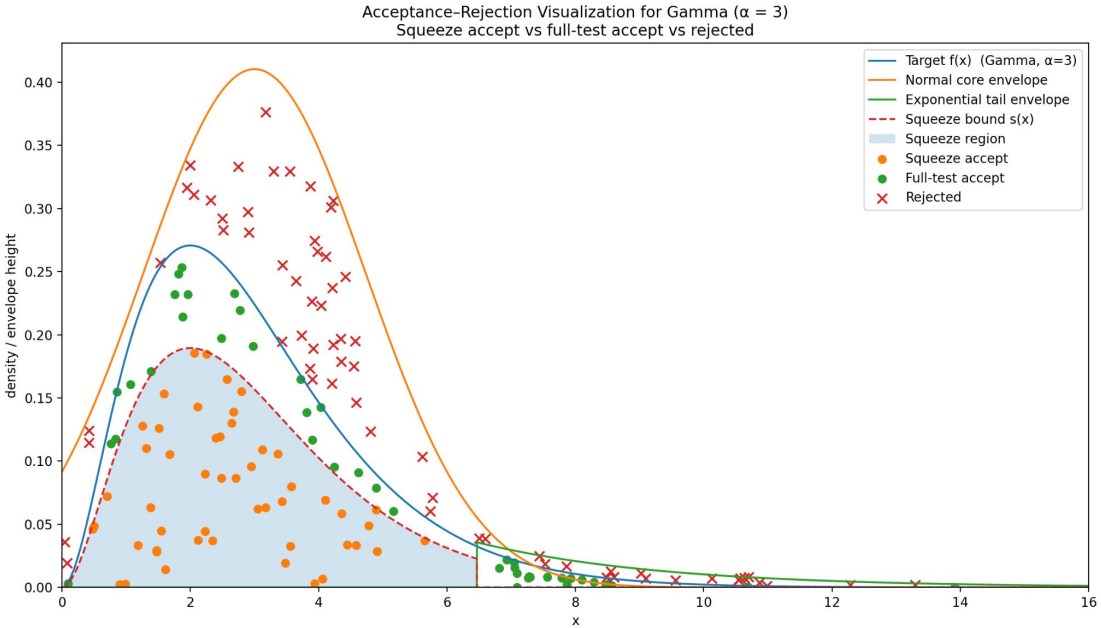
This tradeoff is the core challenge Dieter & Ahrens (1974) solved for the Gamma distribution.

Three Gamma Algorithms

- There is no universal envelope for the Gamma
  - Gamma Distribution takes different shapes as α increases
- Three different algorithms
  - GS ($0 < \alpha \leq 1$)
  - GC ($1 < \alpha \leq 2.53$)
  - GO ($\alpha > 2.53$)

- Why 2.53?
  - For large α, Gamma looks approximately normal
  - The GO method uses a normal-based approximation
  - Only works safely once α is large enough
  - Might accidentally accept values instead of reject and vise versa if not accounted for





Algorithm GS $(0 < a \leq 1)$

Problem: Gamma density has an infinite spike at zero — standard envelopes fail.

Solution: Two-piece piecewise envelope (Lemma 4) — power function near zero, exponential in the tail.

Efficiency: 1.0 – 1.4 trials per sample across all valid a.

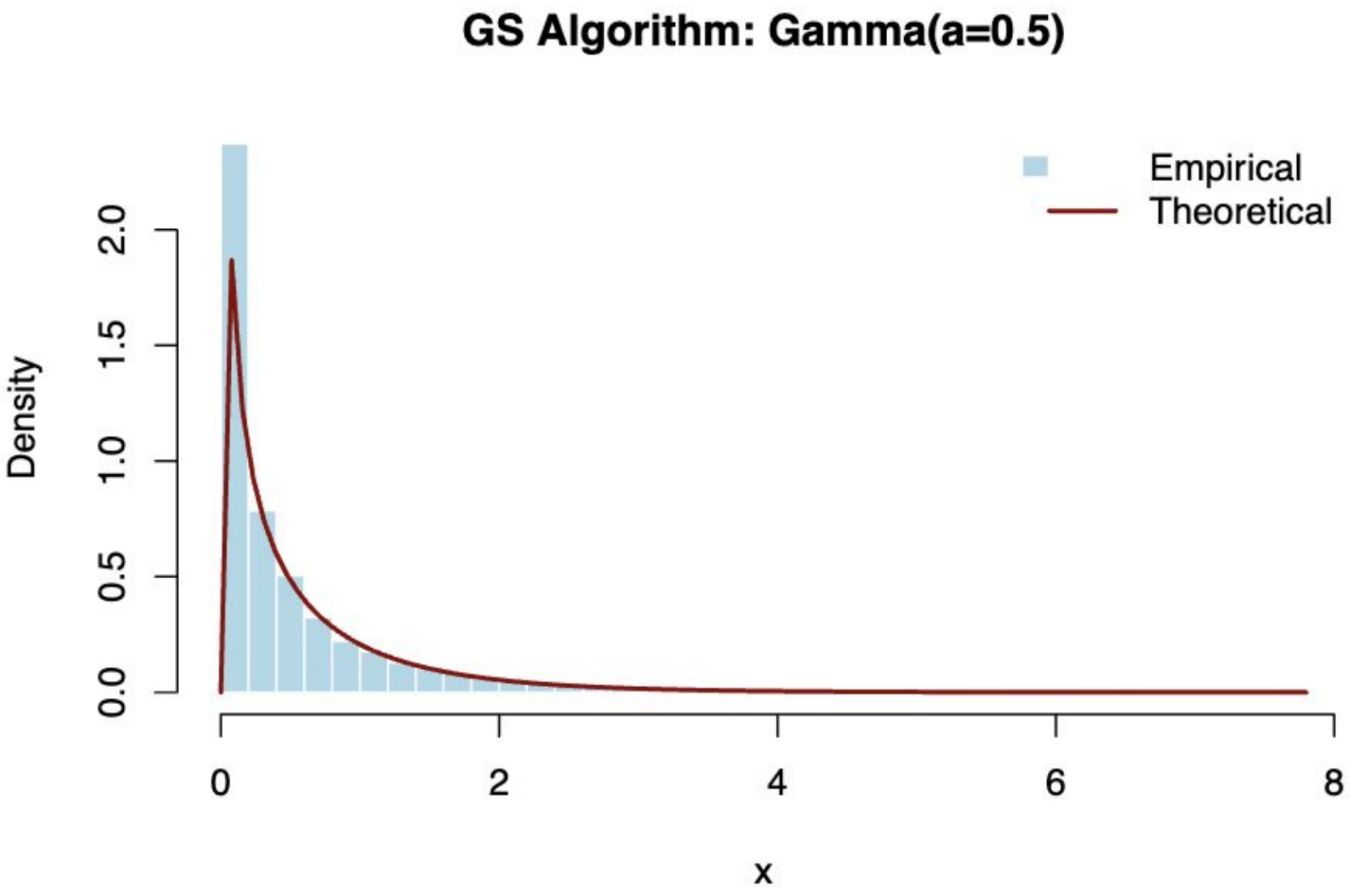
Setup: Set $b \leftarrow (e + a)/e$

Step 1: Generate $u \sim \mathrm{Uniform}(0, 1)$. Set $p \leftarrow bu$.

Step 2: If $p \leq 1$ go to Step 3, otherwise go to Step 4.

Step 3: *(Core region, $x \leq 1$)* Set $x \leftarrow p^{1/a}$. Generate u'. If $\log(u') > -x$ go back to Step 1. Otherwise deliver x.

Step 4: *(Tail region, $x > 1$)* Set $x \leftarrow -\ln\left(\frac{b-p}{a}\right)$. Generate u'. If $\log(u') > (a-1)\log(x)$ go back to Step 1. Otherwise deliver x.


GS Algorithm: Gamma(a=0.5)
Empirical
Theoretical
Density
0.0
0.5
1.0
1.5
2.0
0
2
4
6
8
x

Algorithm GC $(1 < a \leq 2.53)$

Problem: Asymmetric mound shape with a heavy right tail.
Solution: Cauchy envelope — polynomial tails decay at $O(x^{-2})$, thick enough to sit above the Gamma's exponential tail where a Gaussian would fail.
Efficiency: Expected trials decreases from $\pi \approx 3.14$ at $a = 1$ toward $\sqrt{\pi} \approx 1.77$ as $a \to \infty$.

Setup: Set $b \leftarrow a - 1,\ A \leftarrow 2a - 1,\ s \leftarrow \sqrt{A}$

Step 1: Generate $u \sim \mathrm{Uniform}(0, 1)$. Set

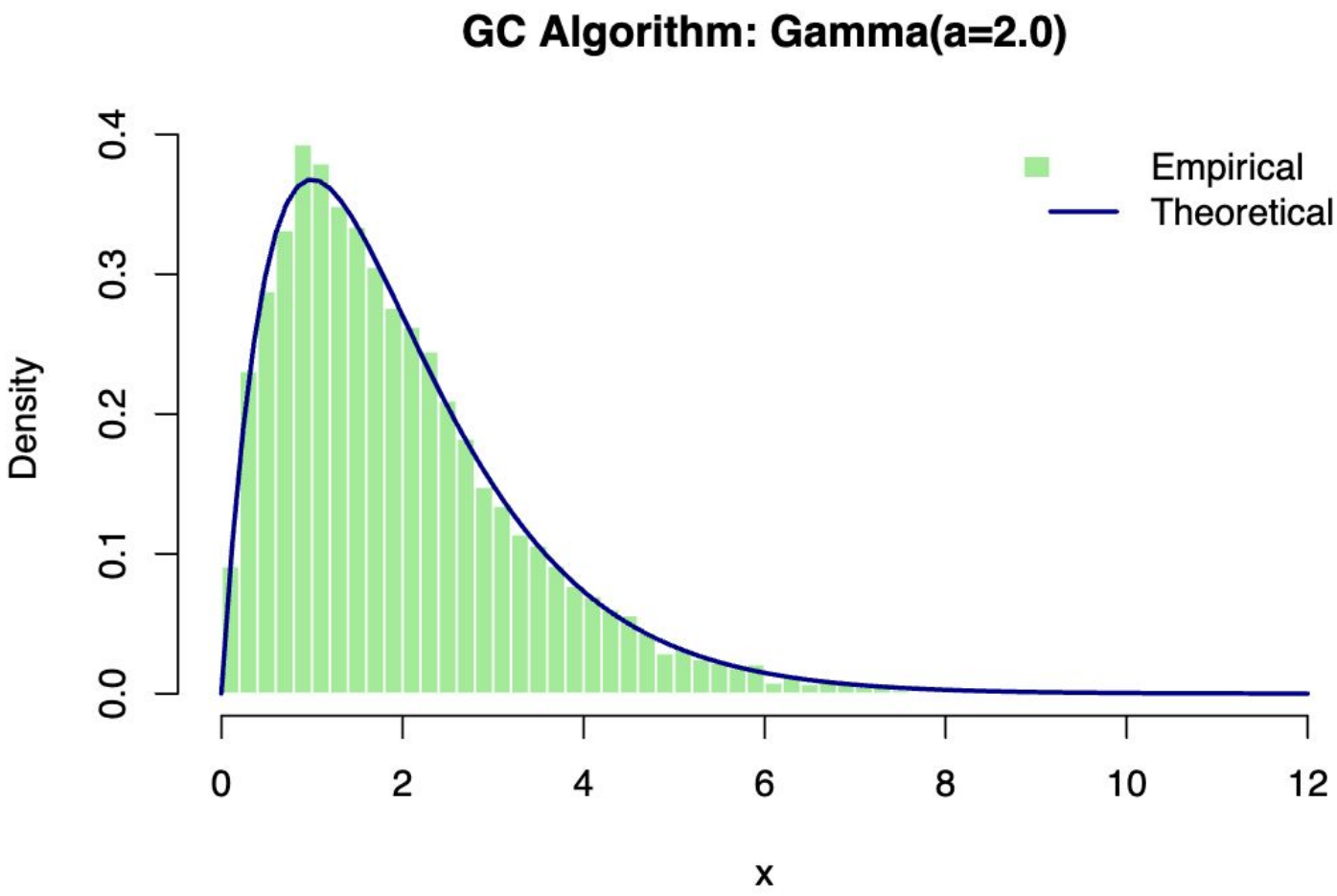
$$t \leftarrow s \tan\left(\pi\left(u - \tfrac{1}{2}\right)\right), \qquad x \leftarrow b + t$$

Step 2: If $x < 0$ go back to Step 1.

Step 3: Generate u'. If

$$\log(u') > b \ln\left(\frac{x}{b}\right) - t + \ln\left(1 + \frac{t^2}{A}\right)$$

go back to Step 1. Otherwise deliver x.


GC Algorithm: Gamma(a=2.0)
Empirical
Theoretical
Density
0.0
0.1
0.2
0.3
0.4
0
2
4
6
8
10
12
x

Algorithm GO ($a > 2.53$)

Problem: For large a, Gamma ≈ Normal — but no Gaussian can fully enclose the right tail.
Solution: Normal-Exponential split with squeeze bounds — resolves $> 99\%$ of proposals without any $\log()$ calls.

Setup: $\mu \leftarrow a - 1, \quad \sigma \leftarrow \sqrt{a + \sqrt{\frac{8}{3}}\sqrt{a}}, \quad W \leftarrow \frac{\sigma^2}{\mu}, \quad d \leftarrow \sqrt{6}\,\sigma, \quad b \leftarrow \mu + d$

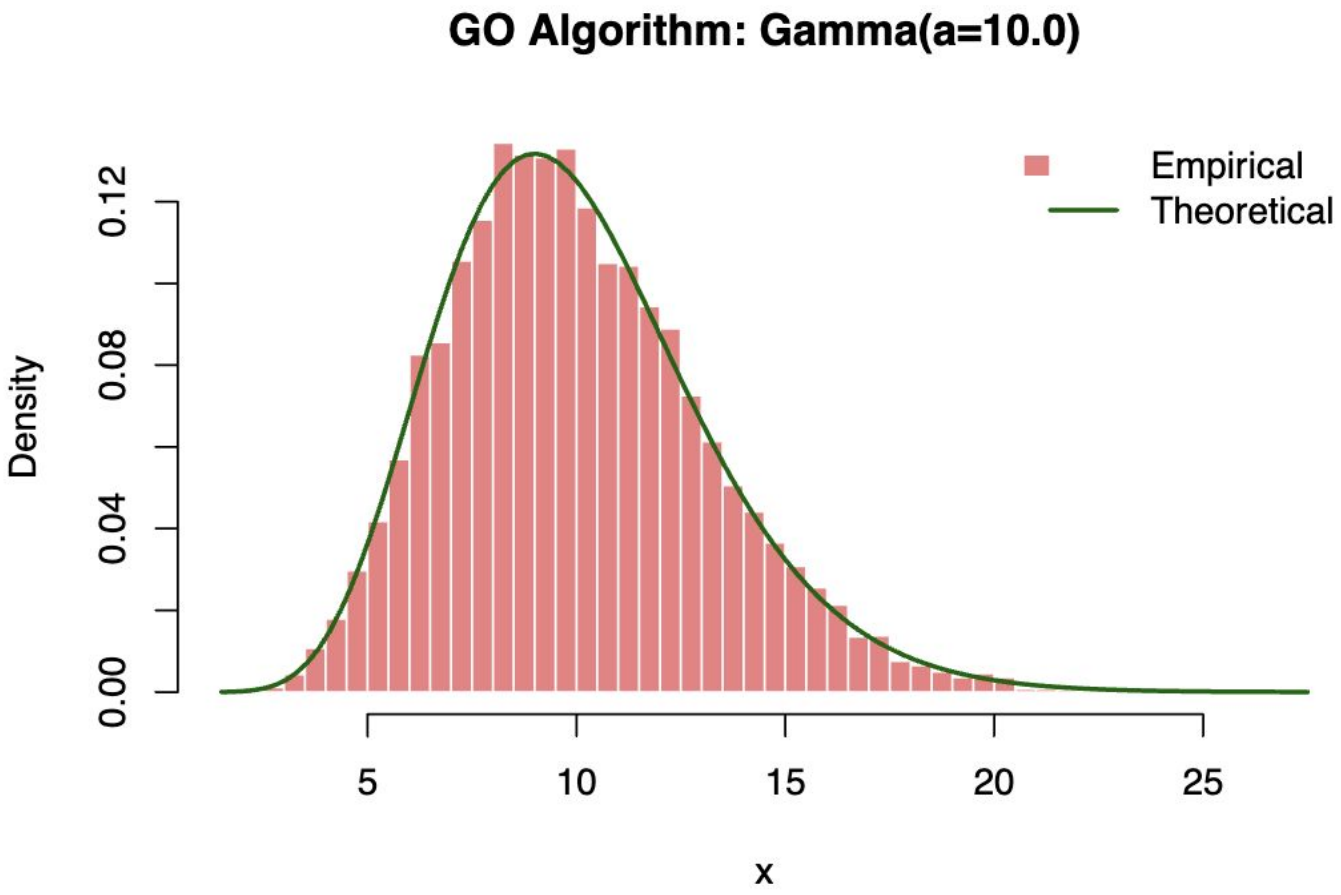
Core region ($u_1 > \beta$): Draw $s \sim N(0,1)$, set $x \leftarrow \mu + \sigma s$

Compute squeeze bound L:

$$L = \begin{cases} 1 - S(W-1) & s \geq 0 \\ 1 - S(W-1) + \dfrac{s^3}{\sqrt{a}}\,W & s < 0 \end{cases}$$

If $u_2 \leq L$ deliver x immediately. Otherwise apply full log test.

Tail region ($u_1 \leq \beta = 0.009572\ldots$): Draw $s_e \sim \mathrm{Exp}(1)$, set $x \leftarrow b(1 + s_e/d)$, apply log test.


GO Algorithm: Gamma(a=10.0)
Empirical
Theoretical
Density
0.00
0.04
0.08
0.12
5
10
15
20
25
x

Using Gamma RNGs to generate a Beta RNG

Beta is not itself a Gamma distribution, but it can be constructed from Gamma random
variables.

Let

$$X \sim \mathrm{Gamma}(\alpha_1, 1), \qquad Y \sim \mathrm{Gamma}(\alpha_2, 1), \qquad \text{independent.}$$

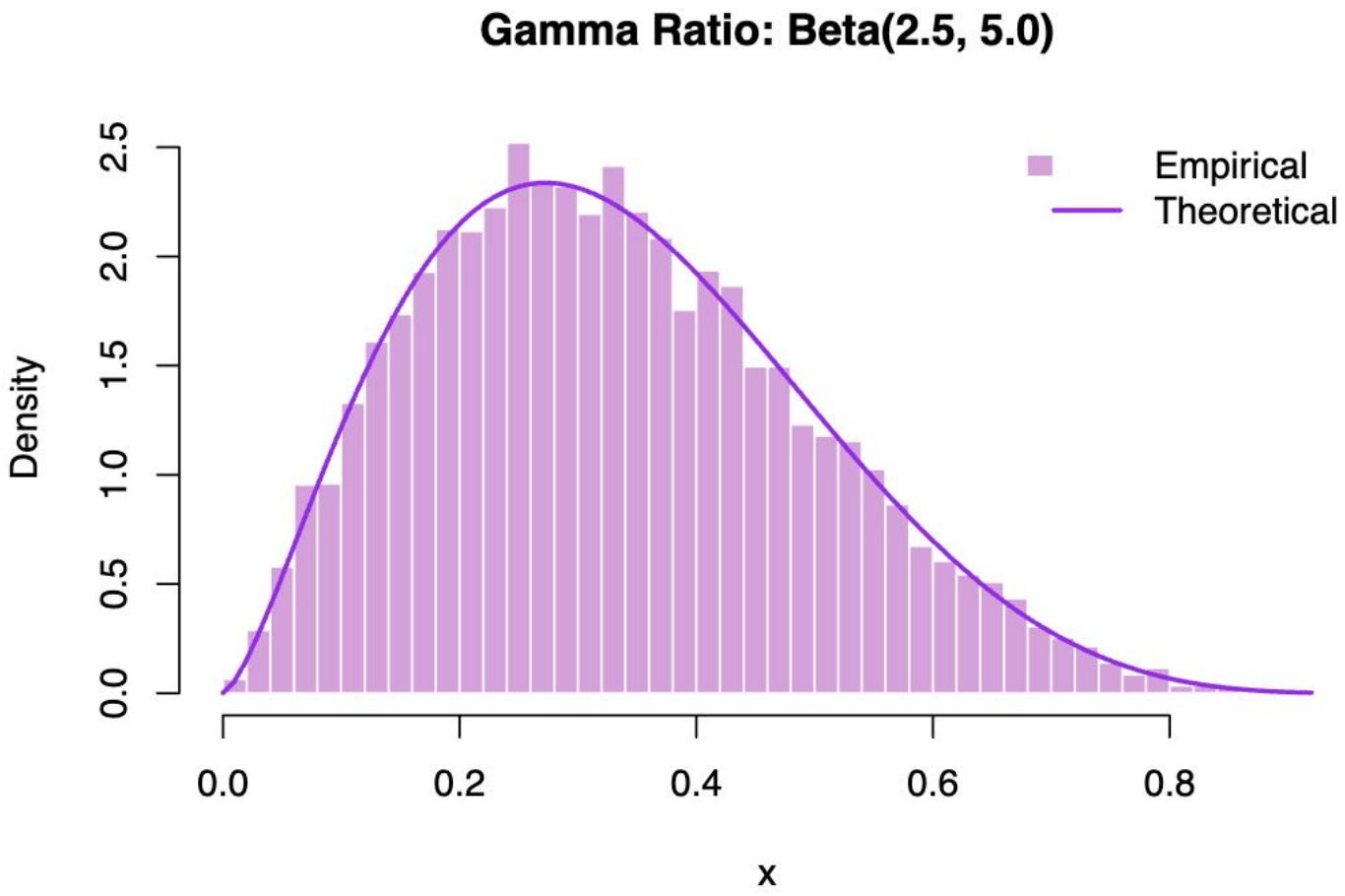
Define

$$Z = \frac{X}{X+Y}.$$

Then

$$Z \sim \mathrm{Beta}(\alpha_1, \alpha_2).$$

Basically, once we have a Gamma RNG, we can generate Beta samples by drawing two
Gammas and normalizing.


Gamma Ratio: Beta(2.5, 5.0)
Empirical
Theoretical
Density
0.0
0.5
1.0
1.5
2.0
2.5
0.0
0.2
0.4
0.6
0.8
x

Verifying our Results: Kolmogorov–Smirnov test

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--- K-S Test Results ---
Algorithm GS (a=0.5) p-value: 0.6150175
Algorithm GC (a=2.0) p-value: 0.6292458
Algorithm GO (a=10.0) p-value: 0.5681361
Gamma-Ratio Beta Generator (a=2.5, b=5.0) p-value: 0.7920064
```

- Measures the maximum vertical distance between the Empirical CDF and Theoretical CDF
- We fail to reject H_0 for all 4 distributions since $p > 0.05$

H_0: The generated samples come from the specified theoretical distribution

$$H_0 : F_n(x) = F(x)$$

H_1: The generated samples do NOT come from the theoretical distribution.

$$H_1 : F_n(x) \neq F(x)$$